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$$f(x,y) = \begin{cases} \dfrac{x^3 - y^3}{x^2 + y^2} & \text{als } (x,y) \neq (0,0) \\ 0 & \text{als } (x,y) = (0,0) \end{cases}$$

continuïteit:

$x = r\cos(\theta)$ en $y = r\sin(\theta)$ :

$\lim\limits_{r\to 0} \dfrac{r^3\cos^3(\theta) - r^3\sin^3(\theta)}{r^2\cos^2(\theta) + r^2\sin^2(\theta)} = \lim\limits_{r\to 0} \dfrac{r^3(\cos^3(\theta) - \sin^3(\theta))}{r^2} = \lim\limits_{r\to 0} r(\cos^3(\theta) - \sin^3(\theta)) = 0$

$\Rightarrow$ de limiet is gelijk aan $f(0,0) = 0$, dus $f$ is continu in $(0,0)$.

partiële afgeleiden:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{\lambda\to 0} \frac{f(\lambda,0) - f(0,0)}{\lambda} = \lim_{\lambda\to 0} \frac{\frac{\lambda^3 - 0}{\lambda^2 + 0} - 0}{\lambda} = \lim_{\lambda\to 0} \frac{\lambda}{\lambda} = 1$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{\lambda\to 0} \frac{f(0,\lambda) - f(0,0)}{\lambda} = \lim_{\lambda\to 0} \frac{\frac{0 - \lambda^3}{0 + \lambda^2} - 0}{\lambda} = \lim_{\lambda\to 0} \frac{-\lambda}{\lambda} = -1$$

$\Rightarrow$ beide partiële afgeleiden bestaan in $(0,0)$.

afleidbaarheid:

stel $\bar{h} = (h_1, h_2)$ en $h_1 \neq h_2 \neq 1$

$$Df(\bar{0},\bar{h}) = \lim_{\lambda\to 0} \frac{f(\lambda h_1, \lambda h_2) - f(0,0)}{\lambda}$$

$$= \lim_{\lambda\to 0} \frac{\frac{\lambda^3 h_1^3 - \lambda^3 h_2^3}{\lambda^2 h_1^2 + \lambda^2 h_2^2} - 0}{\lambda} = \lim_{\lambda\to 0} \frac{\lambda^3(h_1^3 - h_2^3)}{\lambda^3(h_1^2 + h_2^2)} = \frac{h_1^3 - h_2^3}{h_1^2 + h_2^2}$$

$\Rightarrow$ alle richtingsafgeleiden bestaan, dus $f$ is afleidbaar in $(0,0)$.

differentieerbaarheid:

$Df(\bar{0},\bar{h}) \neq Df(\bar{0}) \cdot \bar{h}$

$\dfrac{h_1^3 - h_2^3}{h_1^2 + h_2^2} \neq h_1 - h_2$

$\Rightarrow f$ is niet differentieerbaar in $(0,0)$.

## References